

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Artificial Intelligence (AI) and Data Science) SEMESTER - 5 Winter 2025 (Regular)

Course :Bachelor of Technology (Artificial Intelligence (AI) and Data Science) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTAIPE504D - ADVANCED JAVA

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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- Q1. Objective type questions. (Compulsory Question) 12
- 1 Which container is used as the top-level window in Swing applications?
a) JPanel b) JFrame c) JDialog d) JWindow
 - 2 Which object holds the result of an executed SQL SELECT query?
a) Connection b) Statement c) ResultSet d) DriverManager
 - 3 Which scenario justifies using Vector instead of ArrayList?
a) Faster iteration b) Memory optimization
c) Thread-safe operations without external synchronization
d) Sorted data storage
 - 4 Which statement type is used to execute stored procedures?
a) Statement b) PreparedStatement c) ResultSet d) CallableStatement
 - 5 Which method is used to obtain the URL of the HTML page that loaded the applet?
a) getDocumentBase() b) getCodeBase()
c) getAppletContext() d) getParameter()
 - 6 Which tool is used to generate stub and skeleton classes in traditional RMI?
a) javac b) jar c) javadoc d) rmic
 - 7 Which event class is generated when a key is typed (character input) rather than pressed or released?
a) KeyPressedEvent b) KeyReleasedEvent
c) KeyEvent with KEY_TYPED d) InputEvent
 - 8 Which object acts as a proxy for the remote object at the client side?
a) Skeleton b) Remote reference c) Stub d) Registry
 - 9 Which layout manager arranges components in five regions: North, South, East, West, and Center?
a) FlowLayout b) GridLayout c) CardLayout d) BorderLayout
 - 10 Which feature of PreparedStatement improves performance for repeated SQL execution?
a) Query caching b) Precompilation of SQL statements
c) Automatic transaction handling d) Dynamic query rewriting

- 11 Which layout manager places components one after another in a row?
a) FlowLayout b) BorderLayout c) GridLayout d) BoxLayout
- 12 Which class is used for UDP communication?
a) Socket b) ServerSocket
c) DatagramSocket d) URLConnection
- Q2. Solve the following.
- A) Define an applet. Explain how parameters are passed from an HTML page to an applet with an example. 6
- B) Compare AWT and Swing components, highlighting their differences and providing reasons for those differences. 6
- Q3. Solve the following.
- A) What is hashing in Java collections? Explain the internal working of HashMap. 6
- B) Define component and container in Java GUI programming. Describe the purpose of menus and menu items in GUI applications. 6
- Q4. Solve Any Two of the following.
- A) Define JDBC and list its components. Explain the JDBC architecture with a neat diagram. 6
- B) What is a PreparedStatement? Write a JDBC program using PreparedStatement to insert records into a database. 6
- C) Name the four types of JDBC drivers. Explain the JDBC connection process step-by-step. 6
- Q5. Solve Any Two of the following.
- A) Define Remote Method Invocation. Explain RMI architecture with a neat diagram. 6
- B) You are developing a distributed application where clients can perform arithmetic operations remotely. Design a Java RMI interface for a calculator service that allows clients to perform basic operations like addition, subtraction, multiplication, and division. 6
- C) Explain how RMI handles parameter passing and return values between client and server. 6
- Q6. Solve Any Two of the following.
- A) What is a servlet? Describe the stages of a servlet's lifecycle, accompanied by a labeled diagram. 6
- B) Define socket and server socket. Write a TCP client program that establishes a connection with a server and performs data communication. 6
- C) Write a simple program using a servlet that displays a welcome message. Discuss the differences between servlets and applets with examples. 6

***** End *****